

8 Ans: C

Force constant, $k = \frac{5.0}{0.1} = 50 \text{ N m}^{-1}$

Initial extension $= \frac{F}{k} = \frac{3}{50} = 0.06 \text{ m}$ and final extension $= \frac{F}{k} = \frac{2.5}{50} = 0.05 \text{ m}$

Change in E.P.E. $= \frac{1}{2} k x_f^2 - \frac{1}{2} k x_i^2 = \frac{1}{2} (50) (0.05)^2 - \frac{1}{2} (50) (0.06)^2 = -0.028 \text{ J}$

9 Ans: B